

A Lorentz-invariant background and the state-route constant: a conditional link to the radiation unit

Result, 2026-09-27; revised the same day after an adversarial Fable review. STATE item 2 asked for a physical reason why the constant ζ of the state route (GPT-6 Astra's routes note, floor with $h_* = 2\zeta$; the fifth-postulate note, Theorem B') should be a fixed multiple of the radiation unit h_{rad} of Theorem U (unit-and-indeterminacy note). Stochastic electrodynamics supplies one, conditionally.

1. A stationary random electromagnetic background that is Lorentz invariant has energy $\kappa\omega$ per normal mode, one constant κ with the units of action.
2. For a charged oscillator in equilibrium with it, the *resonant* part of the motion is Gaussian with $\sqrt{\det \bar{\Sigma}} = \kappa$, independent of mass, charge and frequency (Theorem 1). The full momentum variance diverges in the Lorentz-invariant background; a frequency cut-off removes the divergence and breaks the invariance that fixed the spectrum. So the background realizes the restriction of Theorem B' with $\zeta = \kappa$ for band-filtered (resonant) variables.
3. *If* the Planck spectrum, zero-point term included, is derived within stochastic electrodynamics, the thermal quantum is twice the zero-point constant: $h_P = 4\pi\kappa$, where h_P is Planck's constant. With $2\zeta = 2\kappa$ this gives $2\zeta = h_P/(2\pi)$. Relative to Theorem U's h_{rad} , which differs from h_P by the pure number $h_P = (\pi^4/15)^{1/3}h_{\text{rad}}$ stated in the unit note,

$$2\zeta = \gamma h_{\text{rad}}, \quad \gamma = \frac{(\pi^4/15)^{1/3}}{2\pi} \approx 0.297.$$

Every stochastic-electrodynamics derivation of the Planck spectrum is disputed in that literature (§3), and the value $\kappa = \hbar/2$ is then experimental, fixed by matching the zero-point term. The link is therefore conditional on step 3 and restricted to resonant variables by step 2. Astra's closure premise, that records cannot read the background and leave its equilibrium set, stays open. The ingredients are established; the assembly is this note's, with no novelty claimed for the parts.

1. The background

A homogeneous, isotropic, stationary random electromagnetic field whose spectrum is Lorentz invariant has energy density proportional to ω^3 , that is, energy per normal mode linear in frequency, $\kappa\omega$. Boyer states it in the abstract of his derivation (Boyer 1969, abstract: the zero-point spectrum is "linear in frequency", $\frac{1}{2}\hbar\omega$ per normal mode); the random-electrodynamics programme goes back to Marshall (1963) (metadata). Lorentz invariance leaves κ free.

2. Theorem 1: the resonant Gaussian area

Theorem 1. Let a particle of mass M and charge e be bound harmonically at frequency ω_0 with radiation damping, $M\ddot{x} = -M\omega_0^2x + M\tau\ddot{x} + eE_x$, $\tau = 2e^2/(3Mc^3)$, weak damping $\tau\omega_0 \ll 1$, driven by the Gaussian background of §1. The stationary state is Gaussian, and its resonant part (frequencies within a few linewidths of ω_0) has

$$\langle x^2 \rangle_{\text{res}} = \frac{\kappa}{M\omega_0}, \quad \langle p^2 \rangle_{\text{res}} = \kappa M\omega_0, \quad \langle xp + px \rangle = 0, \quad \sqrt{\det \Sigma_{\text{res}}} = \kappa,$$

independent of M , e and ω_0 .

Proof. Linear response to Gaussian forcing is Gaussian. Near resonance the susceptibility satisfies $|\chi|^{-2} \simeq 4\omega_0^2(\omega - \omega_0)^2 + \tau^2\omega_0^6$, whose integral gives $\int |\chi|^2 d\omega \simeq \pi/(2\tau\omega_0^4)$; with the background's spectral density the charge and τ cancel, and the resonant mean energy is the field

energy per mode at ω_0 , $\kappa\omega_0$ (Planck’s resonator relation for a dipole with radiation damping, Planck 1900, metadata), split equally between kinetic and potential terms. Stationarity gives $\langle xp + px \rangle = 0$. (Hand check of the resonant moments by the reviewer.) $\square$

The off-resonant part. In the ω^3 background the momentum variance has a contribution from $\omega_0 \ll \omega \ll 1/\tau$ of relative size about $1/(\pi\tau\omega_0)$, which grows as the damping weakens, and above $1/\tau$ the integrand behaves like $1/(\tau^2\omega)$: the full $\langle p^2 \rangle$ diverges logarithmically. The position moment is safe; the off-resonant part of $\langle x^2 \rangle$ is of relative order $\tau\omega_0 \ln(1/\tau\omega_0)$. This is the known difficulty of the oscillator in stochastic electrodynamics (Goedecke 1983, metadata; Nieuwenhuizen 2019, abstract: only “after introducing a cut-off in the stochastic power spectrum and regularizing the stochastic force” are the integrals dominated by resonance). A cutoff $\Omega_c \ll \sqrt{\pi\omega_0/\tau}$ keeps the resonant answer, and any such cutoff breaks the Lorentz invariance of §1, as Astra’s routes note already observes. Theorem 1 is therefore a statement about band-filtered variables.

The link to Theorem B’. The Gaussian states with $\sqrt{\det \Sigma} \geq \kappa$ form an affine-symplectic-invariant, noise-closed set, and Theorem 1 puts the resonant variables of every harmonically bound body on its lower boundary. The noise direction that Theorem B’ needs is supplied by a forgotten record, which adds momentum variance only (record note, Proposition 1).

3. The radiation unit, conditionally

Boyer derives the Planck law “without the formalism of quantum theory” from this background, classical electrodynamics of dipole oscillators and classical equipartition of the particles’ kinetic energy (Boyer 1969, abstract), by an Einstein–Hopf argument. What such a derivation fixes is a ratio: the thermal part has the form of Planck’s with quantum h_P , and the zero-point part is $\kappa\omega = h_P\omega/(4\pi)$, so $h_P = 4\pi\kappa$; the value $\kappa = \hbar/2$ is then read from experiment.

The derivation is contested. Senatchin argues that “his derivation contains a loop-hole in its argument”, the wall damping making the equilibrium radiation inhomogeneous (arXiv:physics/0105054, passage as reported by the reviewer), citing Jiménez, de la Peña and Brody (1980) (metadata); Boyer’s own historical review maintains the calculation with a modification of the wall damping (arXiv:1711.04179, passage as reported). Nonlinear systems in stochastic electrodynamics are known to depart from the Planck spectrum (reviewer’s recollection, not checked here). So the honest statement is: within stochastic electrodynamics, and conditional on an SED derivation of the Planck spectrum with its zero-point term,

$$2\zeta = \frac{h_P}{2\pi} = \frac{(\pi^4/15)^{1/3}}{2\pi} h_{\text{rad}} \approx 0.297 h_{\text{rad}}.$$

4. Consequence for STATE

STATE item 2 asked for the identification $2\zeta = \gamma h_{\text{rad}}$ and for closure under recording. The identification holds within stochastic electrodynamics, conditionally on its disputed derivation of the Planck spectrum and for resonant variables, with $\gamma = (\pi^4/15)^{1/3}/(2\pi)$. Closure under recording remains the open premise, now joined by the tension between a Lorentz-invariant background and the cutoff that the oscillator’s momentum variance requires.